

TTPA Variant Analysis

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May 11, 2026

Abstract

Missense mutations in the human *TTPA* gene can disrupt the structural and functional properties of α -tocopherol transfer protein (α -TTP), leading to Ataxia with Vitamin E Deficiency (AVED). Due to the conformationally dynamic nature of α -TTP and its role in vitamin E transport, the structural consequences of missense variants may depend on both local interaction environment and conformational state. In this study, computational structural analyses were performed to investigate the effects of selected clinically relevant *TTPA* variants on protein stability, local interaction environments, and conformational sensitivity.

Structure selection used multi-parameter quality assessment across eight *TTPA* X-ray structures, prioritizing structural certainty in selection. Structure 6ZPD (2.24 Å, zero missing residues, 97% Ramachandran favored) was selected, with structural alignment confirming consensus fold representation (RMSD <0.5 Å). Molecular docking of co-crystallized VIV validated the binding cavity (RMSD = 3.7 Å, G = -10.8 kcal/mol) and identified L183 as a direct ligand contact residue - providing structure-based evidence that this variant may disrupt both protein stability and vitamin E binding.

Variants were selected using ClinVar, COSMIC, cBioPortal, and MAVISp annotations and evaluated using MutateX/FoldX calculations, molecular dynamics-derived conformational clustering, and PyMOL-based structural inspection.

Mutation-induced changes in folding free energy ($\Delta\Delta G$) identified several mutation-sensitive positions, particularly H101, E141, L183, and G246, which showed strong destabilization across multiple substitutions. The most destabilizing substitution, H101W, produced $\Delta\Delta G$ values of 31.72 and 34.78 kcal mol⁻¹ in two representative local interaction environments, indicating context-dependent energetic effects. Structural inspection suggested that the destabilization is primarily associated with local steric packing disruption introduced by the larger tryptophan side chain.

Comparison between single-structure and ensemble-based analyses further demonstrated that predicted mutation effects vary across conformational states, with ensemble analyses generally reducing destabilization estimates while preserving major mutation hotspots. Overall, the results indicate that *TTPA* mutation effects are strongly dependent on local structural environment and conformational context, highlighting the importance of integrating energetic, structural, and dynamic analyses for interpreting pathogenic missense variants.

1. INTRODUCTION

The human *TTPA* gene encodes α -tocopherol transfer protein (α -TTP), a SEC14-like lipid-binding protein responsible for the intracellular transport and systemic retention of vitamin E. Structurally, α -TTP contains a deeply buried hydrophobic cavity that sequesters α -tocopherol, requiring conformational transitions between closed and open states to enable ligand exchange rather than passive diffusion [3, 4]. This gating mechanism, coupled to transient membrane interactions and phosphoinositide-mediated docking, defines a transport cycle in which conformation, membrane association, and ligand exchange are tightly linked [8]. Mutations in *TTPA* cause Ataxia with Vitamin E Deficiency (AVED), characterized by impaired vitamin E retention; importantly, pathogenic missense variants perturb multiple functional layers rather than a single mechanism [1, 5]. This multi-state and multi-layered functional behavior complicates the interpretation of missense variants, as their effects cannot be attributed to a single structural or biochemical mechanism. As a result, computational approaches have become essential for systematically evaluating the impact of mutations across different structural and functional contexts.

Computational approaches for missense variant interpretation include evolutionary and structure-based methods. Evolutionary models infer variant effects from sequence conservation and learned representations, enabling large-scale prioritization of deleterious substitutions but primarily capturing pathogenicity propensity rather than mechanism. In contrast, structure-based approaches such as FoldX estimate changes in folding or binding free energy ($\Delta\Delta G$) and allow direct interrogation of mutation effects within a defined structural context [6]. Integrated frameworks such as MAVISp combine structural and evolutionary signals but often compress diverse molecular effects into limited output categories, constraining mechanistic

interpretation [11].

Despite their utility, variant effect predictors show substantial disagreement and context dependency, with performance varying across proteins and functional classes [9]. A central limitation is the reliance on static structural representations, which fail to capture conformational variability in proteins such as α -TTP that undergo transitions between closed, open, and membrane-associated states. As a result, mutation effects may depend strongly on the structural context in which they are evaluated, and predicted $\Delta\Delta G$ values do not necessarily correspond to the causal mechanism of pathogenicity.

Given the multi-layered and dynamic nature of α -TTP function, an integrative approach that combines structure-based mutagenesis with context-dependent structural analysis is required. In this study, mutation effects are evaluated using computational mutagenesis (MutateX/FoldX) to estimate $\Delta\Delta G$ within defined structural environments. To focus on relevant interaction contexts, the oligomeric structure is reduced to representative substructures, enabling the analysis of local inter-chain interactions. This approach is complemented by structural conformations derived from molecular dynamics datasets, allowing the exploration of how mutation effects depend on conformational state.

The objective of this study is to investigate the structural effects of selected *TTPA* missense variants by integrating structure-based mutagenesis and context-dependent analysis. Specifically, we map clinically relevant variants onto the *TTPA* structure, evaluate their effects on local stability and inter-chain interactions using MutateX in a representative structural conformation, and interpret these effects in relation to local structural environments and conformational variability. We hypothesize that mutation effects in *TTPA* are strongly context-dependent and may vary with both local structural environment and conformational state.

To contextualize these interaction environments, a larger oligomeric structure was reduced into representative local substructures for structural inspection. The full local interaction context involving subproteins 6, 8, and 17 is shown in fig. 1, while the selected reduced substructure used to examine local interfaces is shown in fig. 2. These reduced systems preserve relevant inter-chain contacts while simplifying the structural context, making them suitable for evaluating local mutation effects in a protein system that undergoes conformational transitions during vitamin E transport.

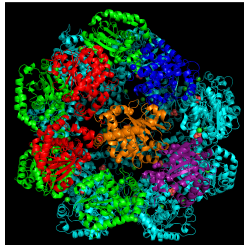


Figure 1. Local TTPA interaction environment containing subproteins 6, 8, and 17. Chain X, corresponding to subprotein 6, is shown in blue; chain Y, corresponding to subprotein 8, is shown in purple; and chain Z, corresponding to subprotein 17, is shown in orange.

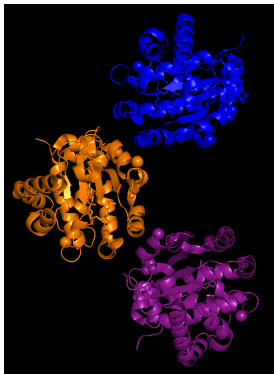


Figure 2. Reduced local 6_8_17 TTPA substructure used to inspect representative inter-chain interaction environments. The same chain colouring is maintained: X/6 in blue, Y/8 in purple, and Z/17 in orange.

The objective of this study is to investigate the structural effects of selected *TTPA* missense variants through structure-based mutagenesis and context-dependent structural analysis. Specifically, clinically relevant variants were mapped onto representative TTPA structural environments and evaluated using MutateX/FoldX to estimate mutation-induced changes in free energy ($\Delta\Delta G$). By comparing mutation effects across representative local interaction contexts, this study aims to explore whether the structural impact of TTPA variants depends on the surrounding interaction environment and conformational state. We hypothesize that mutation effects in TTPA are strongly context-dependent and may vary according to both local structural environment and protein conformation.

2. METHODS

2.1. Gene and variant selection

Missense variants in TTPA were selected based on clinically validated AVED associated substitutions described in foundational genetic and structural studies, including [1] [2]. These works collectively identify pathogenic residues affecting α -TTP stability, ligand binding, or membrane interaction. Based on these sources, the project focused on ten residues: R59, D64, Y74, H101, A120, E141, L183, R192, R221, and G246. These positions span the N-terminal membrane interaction region, the CRAL-TRIO ligand-binding domain, and structurally

constrained helical and C-terminal regions. Their relevance is supported by structural analyses showing that perturbations at these sites can disrupt α -TTP folding, ligand accommodation, or conformational switching. To ensure biological relevance, four complementary databases were used: ClinVar, COSMIC, cBioPortal, and MAVIsP. MAVIsP annotations were based on the integrated structural evolutionary framework described in [10]

2.1.1. ClinVar

ClinVar was used to identify clinically interpreted AVED-associated variants. Several of the selected residues including R59W, A120T, E141K, R192H, and R221W appear in ClinVar as pathogenic or likely pathogenic, consistent with early AVED case studies [1] and structural analyses of disease associated substitutions [5]

2.1.2. COSMIC

COSMIC was queried to determine whether any selected residues appear as somatic substitutions in large tumor sequencing datasets. Although TTPA is not a cancer gene, somatic occurrences at R192 and R221 confirm that these substitutions arise in human populations. This approach aligns with large scale benchmarking of variant effect predictors [9]

2.1.3. cBioPortal

cBioPortal (TCGA + ICGC) was used to cross validate COSMIC findings. Variants appearing in both datasets were considered robust with respect to recurrence across independent sequencing cohorts.

2.1.4. MAVIsP database annotations

MAVIsP was used to map each selected residue onto structural and functional features of α -TTP, including: CRAL-TRIO domain boundaries

- Ligand-binding pocket proximity
- Membrane interaction surfaces
- Secondary structure elements
- Evolutionary conservation.

These annotations were essential for interpreting MutateX/FoldX results and are consistent with structural insights from [3][4]

2.1.5. Prediction of pathogenicity and functional effects

To complement the structure based MutateX/FoldX $\Delta\Delta G$ analysis, pathogenicity and functional impact of the selected TTPA variants were evaluated using two complementary computational tools: AlphaMissense and DeMaSk. These tools provide evolutionary and biochemical perspectives that help interpret whether a substitution is likely to disrupt protein function. AlphaMissense is a deep learning variant effect predictor trained on large evolutionary sequence datasets. It estimates the probability that a missense mutation is pathogenic, following the principles established in large benchmarking studies of variant effect predictors.[9] When applied to the ten selected TTPA residues, AlphaMissense assigned high pathogenicity scores to H101, E141, L183, and G246. These residues are known to be structurally constrained or buried, consistent with earlier structural studies showing their importance for α -TTP stability and ligand binding [3] [5][4]. In contrast, D64 and R192, which are solvent exposed and structurally flexible, received low pathogenicity probabilities. DeMaSk was used to assess whether substitutions at each position are predicted to cause loss of function (LOF) or gain of function (GOF) effects. DeMaSk integrates evolutionary conservation, biochemical similarity, and local structural environment to classify variants as tolerated or deleterious. Consistent with AlphaMissense, DeMaSk identified H101W, E141K, L183P, and G246R as strong LOF associated mutations, reflecting disruption of core packing, ligand-binding

geometry, or backbone flexibility. Exposed residues such as R59, D64, and R192 were predicted to be functionally tolerated, aligning with their low $\Delta\Delta G$ values and minimal structural constraints. Together, AlphaMissense and DeMaSk provide a coherent evolutionary and functional interpretation of the selected variants. Their predictions reinforce the structural findings from MutateX/FoldX and support the conclusion that H101, E141, L183, and G246 represent the most functionally critical and mutation sensitive positions in TTPA.

2.2. Structure Selection and Structural Preparation

The purpose of the structure selection was to identify a single experimental TTPA structure for further downstream analyses. This structure should satisfy a number of requirements in terms of quality and its biological relevance, including ligand binding. Structural selection focused exclusively on Chain A as a representative protomer of the symmetrical 24-mer TTPA assembly for all candidate structure comparisons.

To select a single structure, eight TTPA candidate experimental structures available on RCSB Protein Data Bank (PDB) were compared. Candidates were evaluated according to a number of parameters, see Table 2 (appendix). These include: crystallographic quality (resolution in Å, R-free, and R-work), ligand state (binding of VIV ligand or other), structural completeness (number of unresolved residues (Remark 465), emphasis on cavity residues), Ramachandran analysis (favoured/outlier in %), deposition year, and conformation similarity (RMSD relative to 6ZPD). Local structural order surrounding the ligand-binding cavity was also assessed through B-factor visualization in PyMOL.

The 6ZPD structure was selected as the primary framework for downstream analyses due to a strong aggregate performance across quality parameters, with no critical deficiencies.

2.3. Molecular Docking Methodology

The purpose of the docking analysis of 6ZPD was to assess the accommodation of the biological ligand (VIV; alpha-tocopherol) in the experimentally resolved hydrophobic cavity and whether the this accommodation could be reproduced. All docking analyses were performed only on 6ZPD.

The docking analysis was preceded by structural preparation in PyMOL. This included removing solvent molecules and non-biological buffer components while preserving the experimentally resolved ligand-binding structure.

The docking analysis workflow consisted principally of redocking analysis. To do this, the experimentally resolved VIV ligand was isolated from the deposited structure in PyMOL and prepared for redocking analysis as a separate .mol2 file. This receptor preparation included removal of non-biological solvent and buffer molecules, while preserving cavity-associated structural features.

Docking simulations were performed using the browser-based SwissDock AutoDock Vina program. Localized docking was performed using a $20 \times 20 \times 20$ Å search box centered on the experimentally resolved ligand-binding cavity at coordinates (9.942, -49.717, 24.875 Å). Exhaustiveness was set to 9.

Docking simulations in AutoDock Vina used settings to perform using a static, proximate model of 6ZPD rather than a comprehensive dynamic model. This means that 6ZPD was considered as a rigid receptor with static geometry. The influence of water and other solvents was also not considered, nor were the protein dynamic motion of 6ZPD incorporated.

Resulting docking poses from AutoDock Vina were evaluated qualitatively in PyMOL with the cleaned 6ZPD structure and the experimentally resolved VIV ligand. Docked ligand poses were evaluated against this background to see whether they replicated the experimentally resolved ligand position as well as whether docked ligand

orientation within the pocket was conserved and compatible with pocket geometry.

The docking analysis managed to produce successful redocking of the crystal VIV ligand. The resulting 18 models had a maximum binding affinity to the hydrophobic cavity of $\Delta G = -10.79$ kcal/mol and minimum RMSD to the native VIV ligand of 3.70 Å.

2.4. Ensemble generation and MD processing

2.4.1. Trajectory processing and clustering

Molecular dynamics (MD) trajectories of TTPA were obtained from the ATLAS database. The trajectory was aligned using least-squares fitting on the protein backbone to remove rotational and translational motion. Structural variability was quantified by calculating a pairwise root-mean-square deviation (RMSD) matrix over the trajectory.

Conformational clustering was performed using the GROMOS algorithm with a cutoff of 0.175 nm, corresponding to the average RMSD observed in the trajectory. This procedure identified five distinct conformational clusters, representing different structural states sampled during the simulation.

2.4.2. Selection of representative conformations

To evaluate mutation effects under different structural representations, both single and ensemble-based approaches were employed. The most populated cluster (cluster 1) was selected as a representative structure for single-mode analysis. For ensemble-based analysis, the three most populated clusters were extracted and combined into a multi-model PDB file.

This strategy allowed mutation effects to be evaluated either in a single structural conformation or averaged across multiple conformational states, enabling assessment of structural context dependence.

2.4.3. MutateX calculations (single and ensemble)

Mutation effects were evaluated using MutateX/FoldX to estimate changes in folding free energy ($\Delta\Delta G$) upon amino acid substitution. Calculations were performed in both single-mode (cluster 1) and ensemble mode (clusters 1–3), where ensemble $\Delta\Delta G$ values were averaged across conformations to account for structural variability. Self-scan calculations were performed prior to analysis to assess the reliability of the selected mutation sites.

Solvent accessibility was evaluated using NACCESS, which computes the solvent-accessible surface area (SASA) of residues. The SASA results were compared with the MutateX/FoldX $\Delta\Delta G$ values to assess the relationship between residue exposure and mutational sensitivity. In general, buried residues were more sensitive to mutation, whereas exposed residues were more tolerant to substitution. All calculations were performed on static structural conformations, and therefore primarily capture local energetic effects rather than full conformational dynamics.

2.5. Procedure used to assess the effects of local interactions

For the analyses of the effects of mutations on local interactions, MutateX-FoldX was used. For information regarding MutateX-FoldX, see [13]. Due to the symmetry and octahedral nature of the biological subunit, the biological subunit could be described by the pairs of subproteins, 2₁₇, 6₁₇ and 8₁₇. The subprotein 2₁₇ was located more than 10 Å from the 6 and 8, exceeding the limit for MutateX. Furthermore, as MutateX is not suitable for evaluating interaction over such large spatial separations, the longer interchain interactions that could occur between and/or through e.g 2, 6, 8 and/or 17 are simple not possible to measure in MutateX. As such, 2₁₇ was excluded from the analysis. Consequently, only subprotein 6₁₇ and 8₁₇ were analysed

for their interactions, and any differences in their local interaction due to e.g bio-isomeri, steric effects or chemical interactions.

MAVISp-based classification was used to address when a mutation could be deemed to be, stabilizing, neutral or destabilizing. For information about the classification, see [12].

Table 1. MAVISp-based classification of $\Delta\Delta G$ values into stability categories.

Category	$\Delta\Delta G$ range
Stabilising	$\Delta\Delta G \leq -3$ kcal/mol
Neutral	-2 kcal/mol $< \Delta\Delta G < 2$ kcal/mol
Uncertain	2 kcal/mol $\leq \Delta\Delta G < 3$ kcal/mol or -3 kcal/mol $\leq \Delta\Delta G < -2$ kcal/mol
Destabilising	$\Delta\Delta G \geq 3$ kcal/mol

To further visualise the results and explain any influence from the different types of interactions, Pymol was used to do the 4 Å contact analysis around H101W to visualise the environment of H101W.

3. RESULTS

3.1. Selection and Validation of 6ZPD

Searching the RCSB PDB yielded eight unique results for TTPA experimental structures. The 6ZPD structure was selected as the primary framework downstream analyses due to a strong aggregate performance across quality parameters (see Table 2 in appendix). While it was not the top performing candidate across all parameters, it served as a good compromise with the fewest significant weaknesses. Furthermore, 6ZPD was chosen because it has the fewest contingencies for a meaningful downstream analysis. That is, it has the fewest requirements in terms of pre-processing, and conclusions can be made parsimoniously because of the lack of structural uncertainties. However, of the top five candidate structures, the meaningful differences between structures in terms of impact on analysis is likely fairly small. Figure 3 displays a normalized heat map to visualize the structure selection matrix.

All candidates were resolved with X-ray diffraction. Three structures, 1OIZ, 3W67, and 3W68, were excluded because they either did not include the native VIV ligand, or because they contained other ligands in addition to VIV. Of the remaining five candidate structures, although their unresolved residues were terminally placed, 1OIP and 1R5L were excluded in favour of the remaining three candidates with no unresolved residues. 1R5L had a particularly strong resolution (1.50 Å) and only 11 unresolved terminus residues. However, the lower resolution of the remaining three candidates (2.24-2.42 Å), 5MUE, 5MUG, and 6ZPD, was not sufficiently great to make a large difference in quality.

Of the now three remaining candidate structures, 6ZPD was chosen because it is the most recent model (2020 vs 2017), represents the same conformation as 5MUE and 5MUG (RMSD 0.142-0.199), and has negligible differences in resolution (< 0.18), R-free (< 0.06) and R-work (< 0.09). Ramachandran analysis shows that none of the candidates reviewed here have compromised stereochemistry, with ratios generally around $>96\%$ /(0-1)%. Finally, qualitative B-factor analysis in PyMOL did not suggest any noticeable difference between 5MUE, 5MUG, and 6ZPD.

Albeit most downstream analyses proceeded with 6ZPD, 1R5L would have been a very strong (if not stronger) candidate, and its greater age and missing non-cavity residues may have been very acceptable trade-offs for its greater resolution and better R-free/R-work. Ramachandran did show that 6ZPD (98%/0%) was slightly favoured in its stereochemistry relative to 1R5L (96%/1%). Importantly, the molecular dynamics simulation did use 1R5L, as this was the structure stored in the ATLAS database.

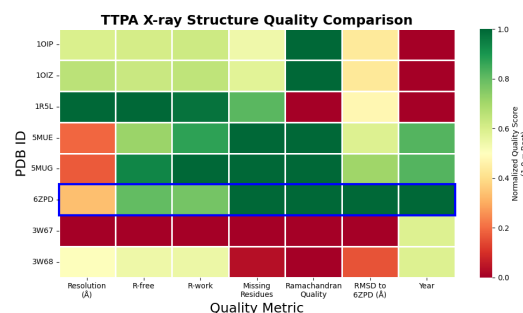


Figure 3. Structure selection framework and scoring. See table 2 for absolute values.

3.2. Redocking successfully validated the 6ZPD Cavity

Redocking with Swissdock AutoDock Vina successfully reproduced binding of the VIV ligand (the alpha-tocopherol analog) to the prepared 6ZPD structure. AutoDock Vina generated 18 models/poses with binding affinities in the range $[-10.79 ; -7.83]$ kcal/mol, with a mean $\pm$ SD of -10.14 ± 0.55 kcal/mol, indicating strong ligand-to-protein interaction consistently across all models.

The top model by affinity scored $\Delta G = -10.79$ kcal/mol and RMSD = 4.28 Å relative to the VIV ligand crystal in the 6ZPD structure from PDB. The model with best redocking geometry yielded RMSD = 3.70 Å. RMSD was moderate across all models. This may reflect the tail of the VIV ligand, which is more unrestricted than the ring segment, leading to more flexibility in total ligand positioning. This became apparent as composite pose inspection in PyMOL showed near-perfect overlap of redocked models, see Figure 4.

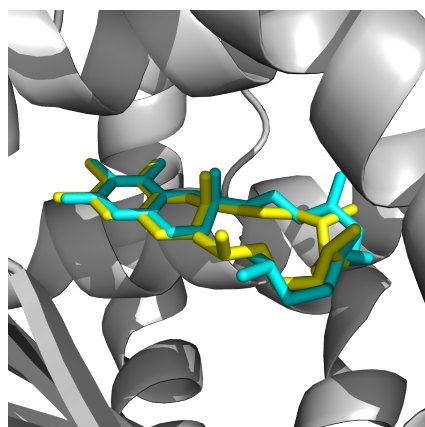
Contact analysis in PyMOL found 17 residues positioned within a 4 Å distance from the docked ligand (making up the VIV-binding interface within the cavity): V132, F133, R134, S136, L137, S140, I154, L156, F158, L160, W163, I179, V182, L183, V191, and F203. Of these, L183 has been identified in this group's work (see later sections on mutation impact) as being associated with severe, destabilizing, and LOF mutations.

3.3. Impact of mutations on Protein Stability

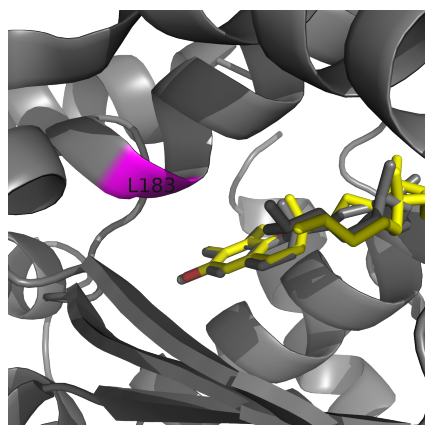
To see how different mutations affect how stable a protein is, we calculated changes in folding energy using a tool called MutateX. This tool is based on another tool called FoldX. We also looked at how much of each residue's exposed to the outside using a tool called NACCESS. Based on what we were asked to do we chose a set of places in the protein to look at: residues 59, 64, 74, 101, 120, 141, 18, 192, 221, and 246.

To understand the changes in folding energy better we looked at these changes along with how much of each residue is exposed to the outside. This exposure is important because it helps us understand how mutations affect the stability of the protein, as shown in Figure 5. One clear example is residue H101. As seen in Figure 5, it is completely buried inside the protein. Mutations at this position make the protein less stable. This suggests that H101 is important for keeping the structure of the protein stable. It likely does this through packing or specific interactions that get disrupted when the residue is changed.

A similar trend is seen for E141 and to an extent, L183. These residues are relatively buried inside the protein. Most mutations to these residues make the protein less stable. In particular changing these residues to proline tends to have a negative effect. This makes sense because proline restricts how flexible the backbone of the protein is. It can disturb the structure of the protein. On the hand residues that are more exposed to the outside tend to tolerate mutations better. For example D64 is very exposed. It shows mutations with little or no effect on stability. R192 behaves similarly. Multiple mutations to



(a) Overlay of VIV in crystallographic (yellow) and redocked ligand poses (cyan; 20 models) within the 6ZPD structure.



(b) PyMOL screenshot showing L183 in binding pocket with crystal VIV ligand and redocked VIV.

Figure 4. Molecular docking validation and mutation analysis.

R192 are. Even make the protein more stable. This suggests that these residues are not tightly constrained by interactions.

Residues that are somewhat exposed such as Y74 and A120 show a mix of effects. Some mutations are tolerated, while others, like proline are more destabilizing. This reflects that these positions contribute to the structure of the protein but still allow some flexibility. An interesting case is G246. Mutations to G246 are consistently very destabilizing. Even though glycine is generally flexible it often plays a role in allowing specific backbone conformations. Replacing it likely introduces strain or restricts necessary flexibility. This explains the changes in folding free energy.

Overall there is a relationship between how exposed a residue is and how sensitive it is to mutations. Buried residues are more sensitive to mutation and are important for stability. Exposed residues are generally more tolerant. Across positions proline substitutions stand out as particularly destabilizing. This highlights the importance of backbone flexibility, in maintaining structure.

3.4. Local interactions and heat-maps

The MutateX/FoldX heat maps for 6₁₇, (fig. 6) and 8₁₇, (fig. 7) show that several of the selected TPPA mutation-sites are vulnerable to amino-acid substitution. In both structures, no mutations resulted in stabilizing effect eg, $0 > \Delta\Delta G$. In contrast, most mutations caused a destabilizing effect eg, $0 < \Delta\Delta G$, indicating predicted destabilization. To classify the effect of stability, the MavisP-classification framework was used, (see table 1 or [12]). It is also shown, that the strongest destabilizing effects were not evenly distributed across all residues,

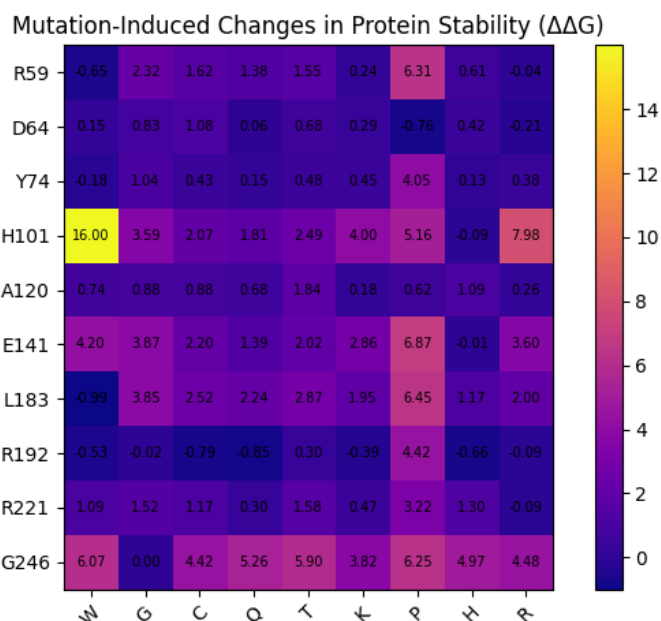


Figure 5. Effect of mutations on protein stability

but concentrated at specific positions and mutation types, with the most apparent ones being H101, E141, L183 and G246. This illustrates that the local structural tolerance differs between mutation sites.

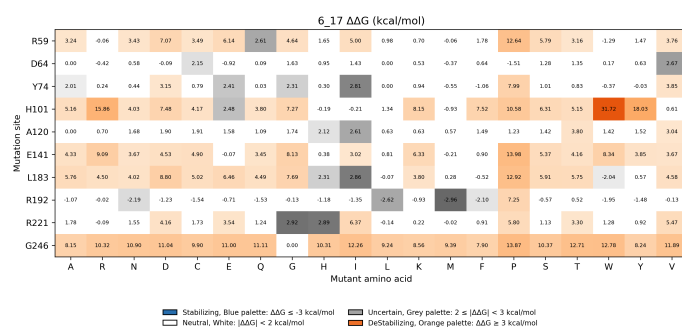


Figure 6. Predicted $\Delta\Delta G$ values for the tested mutations in the 6₁₇ structural context. Higher positive values indicate stronger predicted destabilization.

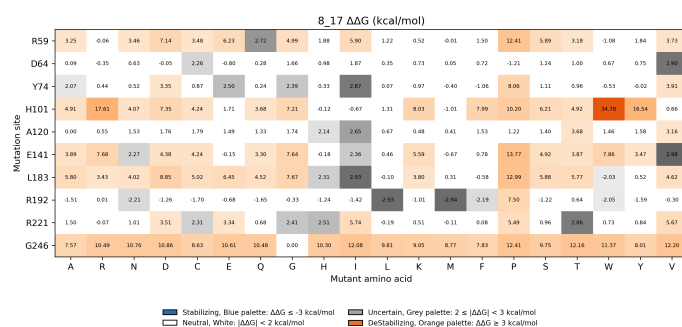


Figure 7. Predicted $\Delta\Delta G$ values for the tested mutations in the 8₁₇ structural context. Higher positive values indicate stronger predicted destabilization.

One clear hotspot was identified at H101W. In both heat maps, replacement of histidine with tryptophan produced the highest predicted destabilization, indicating that this position is particularly sensitive to substitution in both structural contexts. The difference heat map, $\Delta\Delta G(8_{17}) - \Delta\Delta G(6_{17})$ (see fig. 8), shows that H101W does in fact differ between 6₁₇ and 8₁₇. As such, a 4 Å contact analyses and

a visual inspection was done using Pymol,(see fig. 16), in order to evaluate why the predicted effect of this mutation did in fact differ between 6_17 and 8_17.

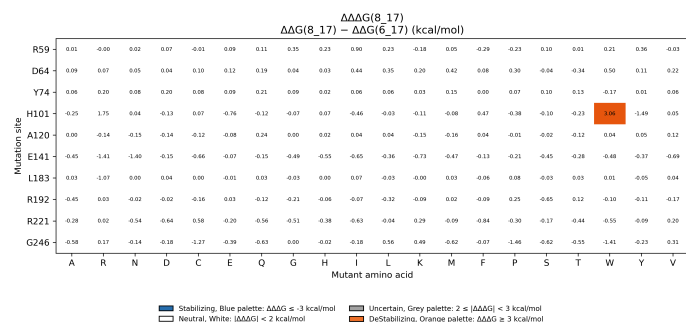


Figure 8. Difference heat map showing the predicted $\Delta\Delta G$ values for 8_17 minus 6_17. Positive values indicate stronger predicted destabilization in 8_17, whereas negative values indicate stronger predicted destabilization in 6_17.

For the H101W mutation,

$$\Delta\Delta G_{6_17}(\text{H101W}) = 31.72 \text{ kcal mol}^{-1},$$

$$\Delta\Delta G_{8_17}(\text{H101W}) = 34.78 \text{ kcal mol}^{-1},$$

Hence for H101W:

$$\Delta\Delta\Delta G_{\text{diff}} = \Delta\Delta G_{8_17} - \Delta\Delta G_{6_17} = 3.06 \text{ kcal mol}^{-1}.$$

3.5. Effect of conformational variability on mutation stability

Clustering of the molecular dynamics trajectory identified five distinct conformational states, with the three most populated clusters selected for ensemble-based analysis. The relative population of each cluster is shown in Fig. 9. The molecular dynamics simulation obtained from the ATLAS database used the 1R5L chain A structure as the starting framework, differing from the 6ZPD structure used in the previous analyses.

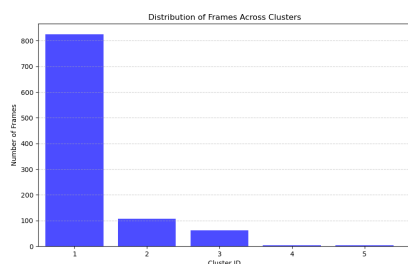


Figure 9. Population distribution of conformational clusters obtained from RMSD-based clustering of the MD trajectory. The three most populated clusters were selected to construct the ensemble used in mutation analysis.

Mutation effects differed between single-mode and ensemble-based analyses. In general, single-mode calculations tended to yield higher $\Delta\Delta G$ values across many positions, indicating stronger apparent destabilization, whereas ensemble-based results showed attenuated effects for several substitutions (Fig. 10). This overall trend suggests that mutation impact is sensitive to structural context and may vary depending on the conformational state considered. Consequently, analyses based on a single static structure may overestimate destabilizing effects for some substitutions.

Despite this overall attenuation, several mutation-sensitive positions identified in the previous analyses remained consistently destabilizing across conformational states. In particular, residues previously associated with strong structural sensitivity, including H101, E141,

L183, and G246, retained elevated $\Delta\Delta G$ values in the ensemble analysis. This suggests that although conformational variability modulates energetic predictions, structurally sensitive mutation hotspots remain robust to changes in structural representation.

In contrast, structurally tolerant positions such as D64 and R192 generally maintained low or near-neutral $\Delta\Delta G$ values across conformations, consistent with the earlier observation that solvent-exposed or structurally permissive regions are less sensitive to substitution. These findings support the interpretation that mutation sensitivity in TTPA depends not only on residue identity but also on structural environment and conformational state.

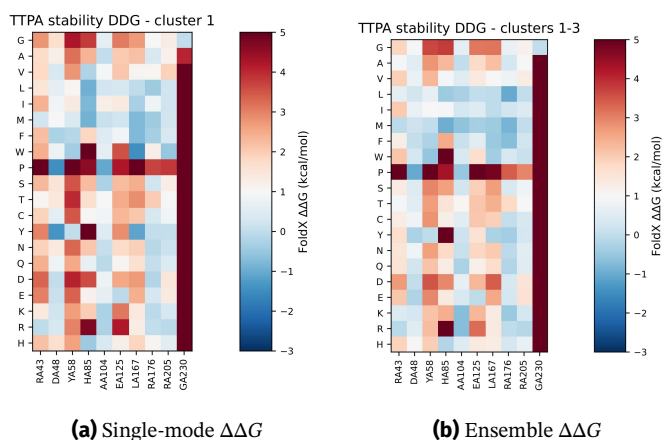


Figure 10. Structural variability and mutation effects across conformational states. (a) Single-mode mutation analysis based on cluster 1. (b) Ensemble-based analysis across clusters 1–3, illustrating how conformational variability influences predicted mutation effects.

One notable example was G246 (Gly230 in the 1R5L numbering), which exhibited consistently high $\Delta\Delta G$ values across both single-mode and ensemble analyses. Interestingly, G246 represented an exception to the general attenuation trend observed in the ensemble analysis. While many positions showed reduced destabilization when multiple conformations were considered, substitutions at G246 frequently remained highly destabilizing and, in several cases, became even more destabilizing in the ensemble-based analysis. For example, aromatic substitutions such as G246F and G246Y increased from approximately 13 kcal mol⁻¹ in the single-mode analysis to more than 20 kcal mol⁻¹ in the ensemble analysis.

To further contextualize this observation, residue-level flexibility from the ATLAS molecular dynamics trajectory was examined. Gly230 displayed moderate RMSF values (0.99 Å) and remained far below the highly mobile terminal regions of the protein, indicating that the destabilizing effect persisted despite local conformational variability (Fig. 11).

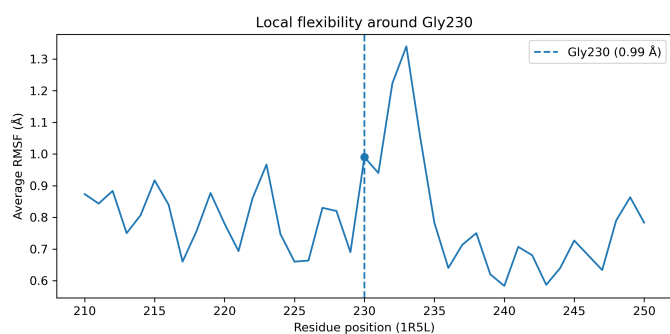


Figure 11. Local residue flexibility around Gly230 (corresponding to G246 in the 6ZPD numbering) obtained from the ATLAS molecular dynamics trajectory. Gly230 is highlighted by the dashed line. The residue exhibits moderate RMSF values relative to neighboring positions and remains far below the highly flexible terminal regions, indicating that the destabilizing effect observed for G246 persists despite local conformational variability.

Inspection of the local structural environment further indicated that G246 is positioned near the boundary of a short helical motif identified in the ATLAS secondary-structure annotation. The presence of consecutive glycine residues (GG) in the surrounding sequence suggests that local backbone compatibility may be important in this region (Fig. 12).



Figure 12. Local structural environment surrounding G246. The residue is highlighted in red, while the surrounding protein surface is shown transparently. G246 is located near the boundary of a short helical motif and adjacent to a consecutive glycine motif (GG), supporting the interpretation that local backbone geometry contributes to the strong destabilizing effect observed upon substitution.

A size-dependent trend was observed for G246 substitutions, where mutations to bulkier hydrophobic and aromatic residues consistently produced the highest destabilization scores. For example, substitutions to phenylalanine, tyrosine, and tryptophan produced substantially larger $\Delta\Delta G$ values than smaller substitutions such as alanine or serine. This pattern is consistent with steric incompatibility at a geometrically restricted position.

Because glycine lacks a side chain beyond hydrogen, it can adopt backbone geometries that may be difficult to preserve with bulkier amino acids, particularly near local secondary-structure transitions. Consequently, substitutions at this position are likely to perturb local geometry and introduce steric strain, explaining why G246 remained strongly destabilizing across both single and ensemble analyses. The persistence of this effect across conformational states further suggests that the observed sensitivity is robust to structural variability rather than an artifact of a single structural representation.

3.6. Pathogenicity Predictions (AlphaMissense + DeMaSk)

Pathogenicity predictions from AlphaMissense and functional classifications from DeMaSk were used to assess whether the selected TPPA variants are likely to disrupt protein function. Both tools showed strong agreement with the MutateX/FoldX $\Delta\Delta G$ profiles. AlphaMissense assigned high pathogenicity scores to H101, E141, L183, and G246, all of which are buried or structurally constrained positions previously shown to be sensitive to perturbation. [3][4]. In contrast, D64 and R192, which are solvent exposed and structurally flexible, received low pathogenicity probabilities. Intermediate residues such as Y74, A120, and R221 showed moderate scores, reflecting mixed structural constraints. DeMaSk produced a similar pattern. Variants H101W, E141K, L183P, and G246R were classified as loss of function (LOF), consistent with disruption of core packing, ligand-binding geometry, or backbone flexibility. Exposed residues (R59, D64, R192) were predicted to be functionally tolerated, aligning with their low $\Delta\Delta G$ values. Following the qualitative interpretation of AlphaMissense and DeMaSk predictions, a combined multi panel figure was generated using the Python workflow,

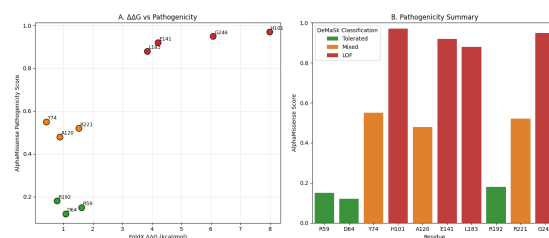


Figure 13. Pathogenicity combined

Figure 13 integrates the estimated pathogenicity scores with the FoldX $\Delta\Delta G$ values to visualise the relationship between predicted functional impact and structural destabilisation. **Panel A** shows a scatter plot of FoldX $\Delta\Delta G$ versus AlphaMissense pathogenicity score for all ten selected residues. A clear trend is observed, residues with high $\Delta\Delta G$ values (H101, E141, L183, G246) also exhibit high AlphaMissense scores, reinforcing their classification as structurally constrained and mutation sensitive positions. Conversely, solvent exposed residues (R59, D64, R192) cluster in the low $\Delta\Delta G$ / low pathogenicity region, consistent with their predicted tolerance. Intermediate residues (Y74, A120, R221) occupy a transitional zone, reflecting mixed structural constraints and moderate predicted impact. **Panel B** summarises the AlphaMissense scores using a bar chart coloured by DeMaSk classification. The same pattern emerges:

LOF-classified variants (H101W, E141K, L183P, G246R) show the highest pathogenicity scores,

Mixed-impact variants (Y74P, A120T, R221W) show intermediate scores, and

Tolerated variants (R59W, D64A, R192H) show the lowest scores.

Together, these panels demonstrate strong agreement between structure-based stability predictions $\Delta\Delta G$ evolutionary based pathogenicity scoring (AlphaMissense), and functional classification (DeMaSk). This multi modal consistency supports the interpretation that buried, structurally constrained residues are highly sensitive to mutation, whereas exposed or flexible residues are functionally permissive.

4. DISCUSSION

4.1. Structure Selection as Trade-off

Protein analyses require the availability of good structures. For consistency and comparability, different analytical methods should be performed on the same, single reference structure. A safe framework for choosing such a structure is to prioritize candidate structures

not simply with good aggregate performance but without any critical deficiencies. As such, all structure selection is to some extent managing trade-offs, particularly with candidate structures of similar performance across quality parameters. With closeness of candidate structures, it can be difficult to objectively prefer one structure over another.

The TTPA structure search produced eight candidates, of which 6ZPD was chosen for downstream analysis, albeit 1R5L was used as supplement. These two candidates, along with 5MUE and 5MUG, show the difficulty of selection. All have parameters that avoid major deficiencies (see Figure 3 and Table 2). In this case, more weight was applied to fully resolved residues, which filtered out 1R5L despite very good resolution, R, and the non-central location of missing residues. 5MUE and 5MUG had similar resolution and R-free/R-work to 6ZPD, but ultimately 6ZPD performed better on the balancing of these parameters. Recency also spoke for 6ZPD, and the 6ZPD structure produced a more competitive Ramachandran scoring of Favoured/Outlier of 97%/0%, compared to 96%/1% of 1R5L, the latter of which was one of the lowest combined scores across all candidates. With its score, choosing the 6ZPD structure further reduced uncertainty in the model.

Ultimately, structural uncertainty was most reduced by prioritizing a recent analysis with no unresolved residues and unfavoured stereochemistry than older, more partial candidates with more unstable performance across parameters.

4.2. Docking Analysis Validates 6ZPD, with Limits

The successful redocking of the VIV ligand to the hydrophobic cavity of the 6ZPD chain A structure validated the computational framework while still showing important limitations. Particularly, RMSD values are moderate and exceed the <2 Å ideal threshold for redocking of smaller molecules such as the VIV ligand. As discussed, this may be congruent with the flexibility of the relatively long branched phytyl tail of the VIV ligand.

However, the fairly tight clustering of all 18 models to within 0.6 Å RMSD show that the Swissdock AutoDock Vina framework did converge on a single overall redocked position. This validates the hydrophobic cavity of the 6ZPD chain A for binding the ligand. Importantly, this current analysis assumes a number of unrealistic parameters for the structure and cavity, namely it being fixed and rigid. A more advanced model considering the induced-fit model of ligand binding of the biological 24-mer TTPA would likely improve quantitative metrics. Indeed, docking models are known to be less compatible with dynamic fitting [2, 7].

4.3. Sterical, orientational and chemical effects of mutation H101W

The results from section 3.4, indicate that H101W is strongly destabilizing in both structural contexts. The magnitude of the effect is consistent with a severe local packing disturbance, rather than a minor side-chain substitution. To explain the difference between $\Delta\Delta G$ of 6₁₇ and 8₁₇, the 4 Å contact analysis in Pymol was used, (see fig. 16). It showed that H101 is surrounded by G99, Y100, G102, I116, Y117, V135, T139, C256, and W259. The high $\Delta\Delta G$ value could be explained by the chemical and steric differences between histidine and tryptophan. Histidine is partially protonated, approx 4% at around 7 pH. Though, since the contact analysis did not find Glu(E) or Asp(D), strong ion-ion interactions are excluded under normal physiological pH conditions, eg. around 7.4 pH. With that in mind, ion-dipole interactions between His and the surrounding polar groups would be possible. Furthermore, His contains a relatively compact imidazole ring, whereas tryptophan contains a much larger indole ring. Replacing H101 with W101 therefore introduces a bulkier aromatic side chain into a local environment while also excluding the potential for any ionic interactions, limiting the strongest interaction to dipole-dipole. With that in mind, large aromatic substitutions are prone

to steric clashes or packing disruption in constrained local environments, is most likely to be the main cause for the substantial overall decrease in stability seen at H101W for both 6₁₇ and 8₁₇.

As seen from, fig. 14, histidine contains a imidazole ring, whereas tryptophan contains a much larger aromatic indole ring, but lacks the ability to do ion-ion interaction under normal physiological pH. These features could explain the destabilization, hence the increased gibbs potential for H101W.

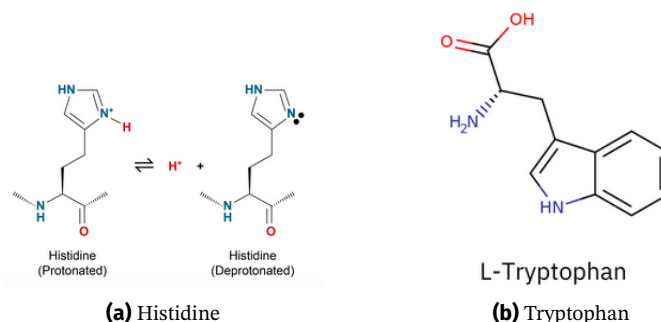


Figure 14. Chemical structures of histidine and tryptophan.



Figure 15. Full view of the locked orientation of H101W

As mentioned, to investigate the difference between the $\Delta\Delta G$ values for 6₁₇ and 8₁₇, PyMOL was used to visualize the H101W mutation in both structural contexts, see (fig. 16). The comparison illustrates that the larger tryptophan side chain is accommodated differently in the two local environments, suggesting that steric packing and side-chain orientation contribute to the higher predicted destabilization in 8₁₇.

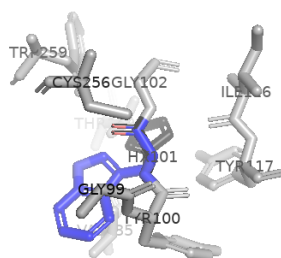
4.4. Influence of conformational context on mutation sensitivity

The ensemble-based analysis suggests that predicted mutation effects in TTPA are sensitive to conformational context. In general, single-structure calculations produced stronger destabilization estimates than ensemble-based predictions, indicating that analyses based on a single static structure may overestimate mutation severity for some substitutions. This observation is particularly relevant for proteins such as α -TTP, whose function depends on conformational transitions associated with ligand exchange and membrane interaction.

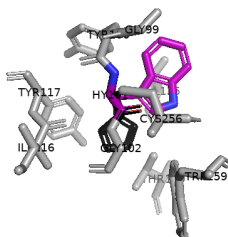
However, not all mutation-sensitive positions behaved similarly across structural states. G246 remained strongly destabilizing across conformations and, for several substitutions, showed increased $\Delta\Delta G$ values in the ensemble analysis. This suggests that some structurally constrained positions are robust to conformational variability. Although residue flexibility alone cannot be interpreted causally, the moderate RMSF values observed for Gly230 (1R5L numbering) indicate that the strong destabilizing signal at G246 is unlikely to be explained solely by local mobility. Instead, local backbone constraints and steric incompatibility introduced by bulkier substitutions likely contribute to the persistent sensitivity of this position.

5. CONCLUSION

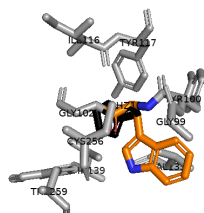
The MutateX local-interaction-analysis, see 3.4, section 2.5 and 4.3, shows that several selected TTPA mutation sites are sensitive to substitution, with many mutations predicted to destabilize the local protein



(a) Chain X, corresponding to subprotein 6: HX101/WX101.



(b) Chain Y, corresponding to subprotein 8: HY101/WY101.



(c) Chain Z, corresponding to subprotein 17: HZ101/WZ101.

Figure 16. PyMOL visualization of the H101W substitution in the local environments of chains X, Y, and Z (orientation shown in fig. 15). Histidine 101 is shown in black, while the H101W substitution is coloured by chain: X/6 (blue), Y/8 (purple), and Z/17 (orange). The figure highlights differences in local side-chain packing introduced by the larger tryptophan residue.

structure. The strongest destabilizing mutation was H101W, which produced $\Delta\Delta G$ values of $31.72 \text{ kcal mol}^{-1}$ in 6_{-17} and $34.78 \text{ kcal mol}^{-1}$ in 8_{-17} . The difference of $3.06 \text{ kcal mol}^{-1}$ indicates that H101W is more destabilizing in the 8_{-17} , which might be explained by steric class, the orientation of W101 compared to H101 and the loss of a potential ion-dipole interaction when substituting H101 with W101.

This is further clarified since a PyMOL inspection showed that H101 is surrounded by a conserved intra-chain local environment consisting mainly of G99, Y100, G102, I116, Y117, V135, T139, C256, and W259. No inter-chain residues were found within $4 \text{ \AA}$ of H101 in chains X, Y, or Z. Therefore, H101W is unlikely to directly disrupt the interface between subproteins. Instead, the destabilization is best interpreted as a local intra-chain packing effect caused by replacing histidine with the much larger tryptophan side chain.

The comparison between 6_{-17} and 8_{-17} indicates that the same mutation can have different energetic effects depending on the local structural context and their orientation. The mutation H101W appears to be less favourably in 8_{-17} than in 6_{-17} , leading to a larger predicted energy penalty. One could see 6_{-17} and 8_{-17} as bio-isomers, steric isomers or simple, distinct sub structures of the same protein, with different local structural environments. However, the $4 \text{ \AA}$ -PyMOL analyses is only a structural illustration and not a full FoldX-relaxed model, as such, the visual interpretation should be treated as supportive rather than definitive. Further analysis using MutateX/FoldX

mutant structures or ensemble-based calculations would be required to determine whether the observed H101W difference is robust across alternative conformations and different analytical tools like e.g. AlphaFold should be used to investigate if mutations affect any long-chain or indirect interactions between the proteins e.g. 2_{-17} , 6_{-17} and 8_{-17} . The stability analysis further supported the importance of local structural environment for mutation sensitivity. Residues with low solvent accessibility, such as H101 and E141, were generally highly sensitive to mutation and showed strongly destabilizing $\Delta\Delta G$ values. In contrast, more exposed residues such as D64 and R192 tolerated mutations better and in some cases even showed slightly stabilizing effects. Together, these results suggest that buried residues play an important role in maintaining the structural stability of TTPA, while exposed residues are less constrained by local packing interactions.

6. DATA AVAILABILITY AND REPRODUCIBILITY

Structure Selection

All relevant data, databanks, code, and commands can be found on the DTU Healthtech Pupil Server under s190867.

Docking

All relevant data, databanks, code, and commands can be found on the DTU Healthtech Pupil Server under s190867.

Stability Analyses

Local Interaction Analyses

The data and code is on the pupil1 healthtech server under s210854, including a guide for the code used.

■ DECLARATION OF USE OF GENERATIVE AI

We have used generative AI tools: yes / no

AI tools we have used:

- ChatGPT

Describe how the tools were used:

What did you use the tool(s) for? Grammar and spelling correction of the report text, clarification of scientific writing, and debugging assistance for scripts and command-line workflows.

At what stage(s) of the process did you use the tool(s)?

During report writing, data analysis, and troubleshooting of computational workflows.

How did you use or incorporate the generated output?

Generated suggestions were manually reviewed and selectively incorporated into the report wording and debugging process. All scientific interpretations, analyses, and conclusions were verified by the authors.

7. AUTHOR CONTRIBUTION

Abstract, introduction, conclusion and other practical aspects of the report: All members did equal work.

Anders Lau Christensen, s210854 see:

"Procedure used to assess the effects of local interactions"(section 2.5), "Local interactions and heat-maps"(section 3.4) and "Sterical, orientational and chemical effects of mutation H101W"(section 4.3).

Peter Munk Adamsen, s203528 see:

"2.4.3. MutateX calculations (single and ensemble)" and "3.3. Impact of mutations on Protein Stability"

Gabriel Luis Dario Loayza Pretel, s252608 see

2.4 Ensemble generation and MD processing, 3.5 Effect of conformational variability on predicted mutation stability, 4.4 Influence of conformational context on mutation sensitivity

Magnus Strange, s190867 see:

"2.2. Structure Selection and Structural Preparation", "2.3. Molecular Docking Methodology", "3.1. Selection and Validation of 6ZPD", "3.2. Redocking successfully validated the 6ZPD Cavity", "3.Y. Molecular Docking Validation", "4.1. Structure Selection as Trade-off", and "4.2. Docking Analysis Validates 6ZPD, with Limits"

Eugenia Kwallu Nilsson s250289 see:

2.1 Gene and variant selection. 2.1.5. Prediction of pathogenicity and functional effects and 3.6 Pathogenicity predictions (AlphaMissense + DeMaSk)

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■ APPENDIX

Table 2. Comparative analysis of available TTPA crystal structures evaluated for docking suitability and structural interpretation, all fetched via RCSB PDB. Where relevant, numbers refer to Chain A.

PDB	Resolution (Å)	R-free	R-work	Method	Ligand State	Unresolved Residues	Ramachandran Favored/Out-lier	Year	RMSD to 6ZPD
1OIP	1.95	0.227	0.191	X-ray	VIV	27	97% / 0%	2003	0.275
1OIZ	1.88	0.226	0.190	X-ray	TRT	25	96% / 1%	2003	0.278
1R5L	1.50	0.209	0.177	X-ray	VIV	11	96% / 1%	2003	0.255
5MUE	2.40	0.222	0.181	X-ray	VIV	0	96% / 0%	2017	0.199
5MUG	2.42	0.212	0.176	X-ray	VIV	0	98% / 0%	2017	0.142
6ZPD	2.24	0.218	0.185	X-ray	VIV	0	97% / 0%	2020	
3W67	2.61	0.255	0.216	X-ray	VIV + PIP	59	96% / 0%	2013	0.485
3W68	2.05	0.230	0.194	X-ray	VIV + PIP	57	98% / 0%	2013	0.409